

1. Create 3-by-3 matrix m , then copy the second and third rows of the matrix twice to create 4-by-3 matrix, then
 - (a) Delete 2^{nd} row.
 - (b) Find sub-matrix of order 2-by-2.
2. Create two matrix of order 3-by-3, add, subtract both matrices-
 - (a) Find multiplication of the both matrix.
 - (b) Find element-wise multiplication of both matrix and see the difference from above multiplication matrix.
3. Given $A = [2, 4, 1; 6, 7, 2; 3, 5, 9]$ and $B = [7, 5, 6; 2, 0, 8; 5, 7, 1]$, find the left division and right division of the matrix,
4. Let $A = [10, 12, 23; 14, 8, 6; 27, 8, 9]$ then perform the scalar operations on the matrix A by scalar quantity $b = 5$.
5. Let $A = [10, 12, 23; 14, 8, 6; 27, 8, 9]$ and $B = [12, 31, 45; 8, 0, -9; 45, 2, 11]$, then find the horizontal concatenation and vertical concatenation of the given matrices.
6. Find the following matrices of the order 3-by-3 :
 - (a) Identity matrix.
 - (b) Zero Matrix.
 - (c) Random number matrix and magic matrix.
7. Create a two dimensional array matrix say A then add a third dimension to A .
8. Perform the relational operators to the given single value -
 - (a) If $a=3$ and $b = 4$
 - (b) If $a= 10$ and $b=5$.
9. Perform the logical operators (AND,OR,NOT) to the given single value -
 - (a) If $a= 10$ and $b = 0$
 - (b) If $a= 2$ and $b = 5$.
10. Perform the logical operator and relational operator to find the relation is 'TRUE' or 'FALSE'-
 - (a) $2+3==5$ OR $3 > 7$.
 - (b) $4+5 == 3$ AND $4 == 4$.
 - (c) NOT $7 > 3$.
11. Define a vector of length 100 between the point 1 and 2, also find the computational time.